

Manhattan Associates, Inc.

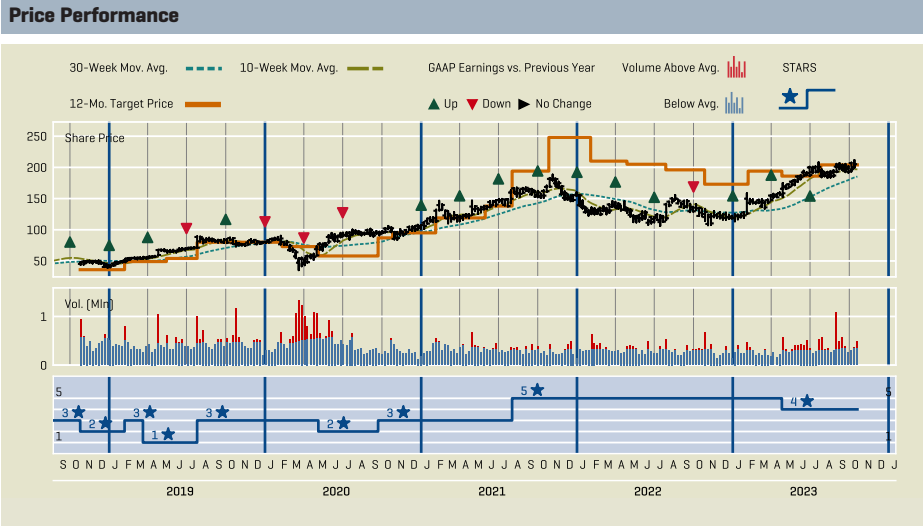
Recommendation **BUY** ★★★★★

Price USD 190.45 [as of market close Oct 20, 2023] **12-Mo. Target Price** USD 204.00 **Report Currency** USD **Investment Style** Mid-Cap Growth

Equity Analyst Janice Quek

GICS Sector Information Technology **Sub-Industry** Application Software **Summary** MANH is a leading supplier of Supply Chain Management (SCM) software and associated maintenance and professional services, especially in the retail business.

Key Stock Statistics (Source: CFRA, S&P Global Market Intelligence (SPGMI), Company Reports)					
52-Wk Range	USD 211.81 - 107.18	Oper.EPS2023E	USD 3.11	Market Capitalization[B]	USD 11.74
Trailing 12-Month EPS	USD 3.15	Oper.EPS2024E	USD 3.52	Yield [%]	N/A
Trailing 12-Month P/E	60.46	P/E on Oper.EPS2023E	61.24	Dividend Rate/Share	N/A
USD 10K Invested 5 Yrs Ago	39,793.0	Common Shares Outstg.[M]	62.00	Trailing 12-Month Dividend	N/A
				Beta	1.53
				3-yr Proj. EPS CAGR[%]	14
				SPGMI's Quality Ranking	B+
				Institutional Ownership [%]	99.0



Analyst's Risk Assessment		
LOW	MEDIUM	HIGH

Our risk assessment reflects rapid changes in the enterprise software industry and increased competition from a large number of emerging cloud-based suppliers with narrower horizontal/functional focus and/or greater vertical industry specialization.

Revenue/Earnings Data					
Revenue (Million USD)					
	1Q	2Q	3Q	4Q	Year
2024	E 239	E 251	E 255	E 260	E 1,005
2023	221	231	E 226	E 215	E 893
2022	179	192	198	198	767
2021	157	166	169	171	664
2020	154	136	150	147	586
2019	148	154	162	153	618

Earnings Per Share (USD)					
	1Q	2Q	3Q	4Q	Year
2024	E 0.75	E 0.87	E 0.98	E 0.92	E 3.52
2023	0.80	0.88	E 0.77	E 0.66	E 3.11
2022	0.60	0.69	0.66	0.81	2.76
2021	0.43	0.61	0.71	0.48	2.23
2020	0.40	0.40	0.51	0.45	1.76
2019	0.41	0.42	0.51	0.40	1.74

Fiscal Year ended Dec 31. EPS Estimates based on CFRA's Operating Earnings; historical earnings are adjusted. In periods where a different currency has been reported, this has been adjusted to match the current quoted currency.

Dividend Data	
No cash dividends have been paid in the last year.	

Source: CFRA, S&P Global Market Intelligence
Past performance is not an indication of future performance and should not be relied upon as such.
Analysis prepared by **Janice Quek** on Jul 27, 2023 01:47 PM ET, when the stock traded at **USD 189.36**.

Highlights	Investment Rationale/Risk
► We anticipate revenue to increase 16.5% in 2023 and 12.5% in 2024, after rising 15.6% in 2022. Solid results in the first half of the year support our upward revision in growth projections for 2023 and 2024, reflecting strong market demand for MANH's range of supply chain and omni-channel solutions. Notwithstanding, the macroeconomic environment remains challenging, resulting in some large deal slippage during the quarter. A small proportion of its clients have filed for bankruptcy, adversely impacting growth in MANH's Remaining Performance Obligations (RPO) in Q2 (about \$15M in lost business); RPO rose 38% Y/Y to \$1.24B during the quarter, compared to 84% Y/Y in Q2 2022. ► We project gross margins at 56% in 2023 and 56% in 2024, compared to 56.2% in 2022. We forecast adjusted operating margins at 27.5% in 2023 and 28% in 2024 [versus 27.6% in 2022] as the benefits of operating leverage are partially offset by the reduction in license and maintenance revenue. ► MANH reported free cash flow of \$39M in Q2 2023, a margin of 17%, contracting from 27% in the prior year. The company maintains a strong balance sheet with \$153M in cash and short-term investments and no long-term debt.	► We maintain our Buy recommendation on shares of MANH on the company's product leadership in warehouse management, transportation management, omnichannel systems, and growing digitization of the industry, driving demand for MANH's software solutions, and fueling services growth downstream. The shift to cloud-delivered software has generated strong interest from existing customers, while contributing to new deal lands; approximately 25% of new cloud bookings are from net new logos. Use cases of MANH's solutions are also growing, while emerging product areas such as its Point-of-Sales (POS) systems are gaining traction. In the medium term, we also see growth upside with AI, as the technology brings in new applications and tools for its customers. ► Risks to our recommendation and target price include deteriorating macro conditions forcing customers to cut IT transformation budgets or defer project spending, less successful cross-sell/upsell motion from limited traction of new solutions, competitive pressures from peers, and a slower pace of product innovation. ► Our target price of \$204 is based on a P/E of 67x our NTM EPS estimate of \$3.05, in line with its three-year average on expanding margins.



Manhattan Associates, Inc.

Business Summary Jul 27, 2023

CORPORATE OVERVIEW. Manhattan Associates, Inc. [MANH] is a leading global provider of technology-based solutions that help companies improve the effectiveness and the efficiencies of their supply chain by coordinating actions of manufacturers, suppliers, distributors, retailers, transportation providers, and consumers. Key benefits of the company's solutions include lower inventory levels, higher order fill rates, reduced transportation costs, and improved labor productivity. MANH offers comprehensive solutions that include software, hardware, and professional services such as design, configuration, implementation, product assessment, training, and customer support.

The company offers two software solution portfolios: MANHATTAN SCOPE [Supply Chain Optimization from Planning through Execution] and MANHATTAN SCALE [Supply Chain Architected for Logistics Execution]. SCOPE utilizes a service-oriented architecture, common data model, and extensive collaborative tools. It is comprised of five product suites: Planning and Forecasting, which helps companies plan their inventory; Inventory Optimization, which helps companies regulate, maintain, and deploy inventory; Order Lifecycle Management, which helps companies to track customers' orders; Transportation Lifecycle Management, which improves the speed, accuracy, and cost of transporting goods; and Distribution Management, which improves the management of goods and labor during distribution operations. MANH also provides analytical tools that provide companies insight and intelligence into the supply chain, such as the consequences of lower product demand and a shipping accident.

SCALE is a logistics execution solution built on the Microsoft NET platform. It is targeted toward small- and medium-sized companies that are focused on improving their trading, shipping, and storage operations.

CORPORATE STRATEGY. MANH originally started as a warehouse management software company that targeted the retail industry, in particular in the soft-line retail sector. The company has broadened its software solutions through internal development and acquisitions. A key acquisition was the purchase of Evant for \$50 million in cash in 2005. We think this acquisition broadened MANH's product footprint into the supply chain planning market segment, which complemented its supply chain execution solutions. Despite this acquisition, we think MANH has a competitive disadvantage versus enterprise resource planning (ERP) software vendors, which provide broader software solutions.

Another component of MANH's growth strategy is to expand international sales. In the past few years, the company has added employees in Asia to increase its presence in that region. However, progress has been slow. Asia Pacific revenues were 5.2% of total sales in 2018, and have remained around that level for a number of years. The company derived most of its revenues from the Americas, which accounted for 80% of total revenues in 2018. Revenues from Europe, the Middle East, and Africa [EMEA] represented 15% of 2018 revenues.

The company competes against a variety of competitors, including internal developed solutions, ERP vendors such as Oracle and SAP, and supply chain execution vendors, including RedPrairie, SSA Global Technologies, and Highjump Software. The number of vendors has declined recently as a result of intense competition and industry consolidation.

FINANCIAL TRENDS. Following a deceleration from 19% and 13% Y/Y growth in 2014 and 2015, respectively, revenue flattened to a 1% compound annual rate from 2015 through 2020. However, MANH's Manhattan Active cloud platform and associated cloud apps have gained strong early traction, leading to a revenue growth acceleration to 13% Y/Y growth in 2021 and 15% in 2022. In our view, MANH's cloud solutions will continue to grow as the cloud growth tailwind picks up momentum.

MANH reported total revenue of \$231 million in Q2 2023, an increase of 20.4% Y/Y from the prior year quarter, continuing its transition to a cloud subscription model. Excluding the impact of the transition, total revenue would have been up 27% Y/Y. Cloud subscription sales is about 58% of total software revenue [cloud, license, and maintenance revenue]. Software license sales fell 27.5% Y/Y in Q2 accordingly. Margins remained strong — adjusted operating margin expanded 210 bps from Q2 2022. while free cash flow margin [17.6%] was down relative to the prior year [27.5%]. We see higher operating leverage as opex as a percentage of total revenue fell when compared to the same quarter of 2022.

MANH has a solid balance sheet ending the quarter with \$153 million of cash and short-term investments, and no debt.

Corporate information

Investor contact

N/A [770 955 7070]

Office

2300 Windy Ridge Parkway, Tenth Floor, Atlanta, Georgia, 30339

Telephone

770 955 7070

Fax

N/A

Website

www.manh.com

Officers

Executive VP, CFO & Treasurer

D. B. Story

Senior VP, Global Corporate Controller & Chief Accounting Officer

L. C. Pinne

Executive VP & CTO

S. Siotia

President, CEO & Director

E. Capel

Senior VP, Chief Legal Officer & Secretary

B. S. Richards

Independent Chairman

J. J. Huntz

Board Members

C. E. Moran

J. J. Huntz

D. Raghavan

K. A. Kuryea

E. Capel

L. T. Hollembaek

E. I. Eger

T. E. Noonan

Domicile

Georgia

Auditor

Ernst & Young LLP

Founded

1990

Employees

4,500

Stockholders

14



Manhattan Associates, Inc.

Quantitative Evaluations						Expanded Ratio Analysis				
Fair Value Rank	1	2	3	4	5		2022	2021	2020	2019
LOWEST HIGHEST Based on CFRA's proprietary quantitative model, stocks are ranked from most overvalued (1) to most undervalued (5).						Price/Sales	10.04	15.07	11.54	8.40
Fair Value Calculation USD 116.26 Analysis of the stock's current worth, based on CFRA's proprietary quantitative model suggests that MANH is overvalued by USD 74.19 or 38.96%						Price/EBITDA	48.30	70.31	55.01	41.90
						Price/Pretax Income	48.68	74.60	59.47	44.73
						P/E Ratio	43.99	69.73	59.76	45.83
						Avg. Diluted Shares Outstg. [M]	63.41	64.32	64.33	65.10
						Figures based on fiscal year-end price				
Volatility	LOW		AVERAGE		HIGH	Key Growth Rates and Averages				
Technical Evaluation	NEUTRAL	Since August, 2023, the technical indicators for MANH have been NEUTRAL"				Past Growth Rate [%]				
Insider Activity						1 Year				
						3 Years				
						5 Years				
						Ratio Analysis (Annual Avg.)				
						Net Margin [%]				
						% LT Debt to Capitalization				
						Return on Equity [%]				

Company Financials Fiscal year ending Dec 31										
Per Share Data (USD)	2022	2021	2020	2019	2018	2017	2016	2015	2014	2013
Tangible Book Value	2.65	2.98	2.47	1.26	1.31	1.66	1.53	1.83	1.62	1.56
Free Cash Flow	2.76	2.86	2.17	2.05	1.96	2.28	1.85	1.48	1.13	1.10
Earnings	2.03	1.72	1.36	1.32	1.58	1.68	1.72	1.40	1.08	0.86
Earnings (Normalized)	2.76	2.23	1.76	1.74	1.79	1.87	1.87	1.52	1.16	0.92
Dividends	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Payout Ratio [%]	NM	NM	NM	NM	NM	NM	NM	NM	NM	NM
Prices: High	157.90	188.52	108.07	89.52	62.39	55.74	68.57	77.75	43.20	31.20
Prices: Low	106.02	101.18	35.20	39.85	39.01	40.75	44.14	37.81	28.24	14.74
P/E Ratio: High	57.20	84.50	61.40	51.50	34.90	29.80	36.70	51.20	37.20	33.90
P/E Ratio: Low	38.40	45.40	20.00	22.90	21.80	21.80	23.60	24.90	24.30	16.00
Income Statement Analysis (Million USD)										
Revenue	767.00	664.00	586.00	618.00	559.00	595.00	605.00	556.00	492.00	415.00
Operating Income	153.00	134.00	114.00	116.00	134.00	189.00	194.00	161.00	127.00	101.00
Depreciation + Amortization	7.00	8.00	9.00	8.00	9.00	9.00	9.00	8.00	6.00	6.00
Interest Expense	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Pretax Income	158.00	134.00	114.00	116.00	136.00	185.00	196.00	163.00	128.00	103.00
Effective Tax Rate	18.40	17.60	23.30	26.10	23.20	37.00	36.60	36.50	35.90	34.70
Net Income	129.00	110.00	87.00	86.00	105.00	116.00	124.00	103.00	82.00	67.00
Net Income (Normalized)	98.80	83.80	71.10	72.50	85.10	117.30	122.60	101.80	80.00	64.40
Balance Sheet and Other Financial Data (Million USD)										
Cash	225.00	264.00	205.00	111.00	101.00	126.00	96.00	129.00	124.00	133.00
Current Assets	415.00	408.00	334.00	232.00	215.00	228.00	207.00	237.00	230.00	219.00
Total Assets	570.20	539.70	465.40	372.30	307.20	315.00	297.10	332.20	318.20	297.80
Current Liabilities	316.00	249.00	197.00	182.00	145.00	124.00	118.00	127.00	123.00	104.00
Long Term Debt	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Total Capital	247.00	274.00	247.00	175.00	147.00	175.00	169.00	195.00	182.00	182.00
Capital Expenditures	7.00	4.00	3.00	15.00	7.00	6.00	7.00	11.00	9.00	5.00
Cash from Operations	180.00	185.00	141.00	147.00	137.00	164.00	139.00	120.00	94.00	89.00
Current Ratio	1.32	1.64	1.70	1.28	1.48	1.84	1.76	1.86	1.87	2.10
% Long Term Debt of Capitalization	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
% Net Income of Revenue	16.80	16.60	14.90	13.90	18.70	19.60	20.50	18.60	16.70	16.20
% Return on Assets	17.20	16.71	17.02	21.33	26.90	38.51	38.59	31.03	25.80	22.62
% Return on Equity	54.00	47.10	48.30	59.30	65.00	67.70	68.10	54.80	45.10	39.20

Source: S&P Global Market Intelligence. Data may be preliminary or restated; before results of discontinued operations/special items. Per share data adjusted for stock dividends; EPS diluted. E-Estimated. NA-Not Available. NM-Not Meaningful. NR-Not Ranked. UR-Under Review.

Manhattan Associates, Inc.

Sub-Industry Outlook

CFRA has a positive fundamental outlook for the S&P 500 Application Software sub-industry, driven by secular digital transformation and cloud migration, albeit negatively impacted in the short term by a slower pace of spending on IT projects. Recent geopolitical and macroeconomic developments have raised expectations that the Federal Reserve could lean toward higher for longer rate hikes. However, inflation growth also appears to be weakening, and we believe the pressure to execute a successful balancing act to keep the U.S. economy from falling into a recession will eventually drive some rate cuts in 2024.

Software spending continued to face headwinds in Q2 as organizations maintained course on keeping budgets tight and reducing discretionary spending. We have observed this behavior for almost three quarters now. While margins for software firms have benefited tremendously, we believe further cuts will be more limited going forward [barring any severe economic shocks in the near term] without hurting productivity and operational goals. Hence, we think demand weakness is stabilizing, and the likelihood that growth compression has “bottomed” is increasing. More observations of this “stabilization” from management in Q2 reinforce our view, and consequently, we expect that we could start to hear of sales momentum improving in Q3 and Q4.

The pace of generative artificial intelligence (AI) developments across the technology sector is also an encouraging sign. Cloud providers are ramping investments in infrastructure hardware to build the necessary computing power to support generative AI workloads, while launching a range of applications and cloud-based services to drive AI model building, training, and application development. For example, Amazon launched Amazon Bedrock, offering a variety of foundational models for users to customize with their data, as well as build and scale their generative AI applications. Larger application software firms also announced new generative AI-based tools and features to augment the functionality of their

current solutions, such as Salesforce with the release of Einstein GPT, which creates personalized content for users in their customer engagement materials. We believe that generative AI tools and software applications will proliferate in the months ahead as cloud providers and software vendors invest to capitalize on current market interest for generative AI products. We also expect that as these products become available, there will be plans to monetize them either through bundling them with higher subscription tiers to drive upgrades, or through an upcharge as an add-on feature. In the near to medium term, we think generative AI will drive the next phase of digital transformation as another source of revenue for the software industry.

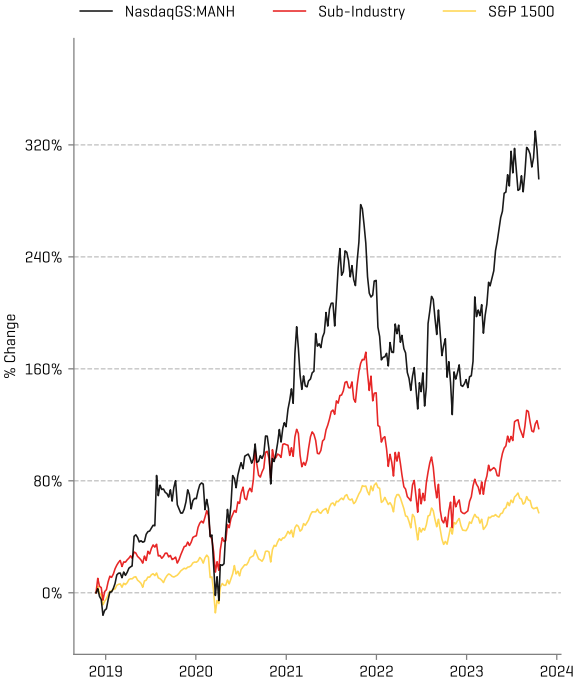
Consensus projections for software align with what we believe are the drivers and impediments to growth in 2023 and 2024. Software companies in the S&P 500 are expected to grow revenue at 9.6% in 2023, down from 12.7% in 2022. However, growth is forecasted to accelerate to 11.9% in 2024, as organizations return to their digital transformation projects, and look at ways to leverage new generative AI solutions in the marketplace. Spending on generative AI can be costly, however, and we expect some impact to earnings, offset by cost savings and efficiency gains in 2023.

/ Janice Quek

Industry Performance

GICS Sector: Information Technology
Sub-Industry: Application Software

Based on S&P 1500 Indexes
Five-Year market price performance through Oct 21, 2023



NOTE: A sector chart appears when the sub-industry does not have sufficient historical index data.
All Sector & Sub-Industry information is based on the Global Industry Classification Standard [GICS].
Past performance is not an indication of future performance and should not be relied upon as such.
Source: CFRA, S&P Global Market Intelligence

Sub-Industry: Application Software Peer Group*: Application Software												
Peer Group	Stock Symbol	Exchange	Currency	Recent Stock Price	Stk. Mkt. Cap. [M]	30-Day Price Chg. [%]	1-Year Price Chg. [%]	P/E Ratio	Fair Value Calc.	Yield [%]	Return on Equity [%]	LTD to Cap [%]
Manhattan Associates, Inc.	MANH	NasdaqGS	USD	196.26	12,103.0	-3.1	54.6	62.0	116.26	N/A	77.0	N/A
AppLovin Corporation	APP	NasdaqGS	USD	38.54	13,404.0	-3.0	121.0	716.0	N/A	N/A	1.2	66.2
Aspen Technology, Inc.	AZPN	NasdaqGS	USD	189.60	12,107.0	-6.2	-23.8	NM	N/A	N/A	-0.8	N/A
BILL Holdings, Inc.	BILL	NYSE	USD	102.50	10,927.0	-1.2	-18.9	NM	N/A	N/A	-5.5	28.4
Confluent, Inc.	CFLT	NasdaqGS	USD	29.82	9,018.0	-6.2	27.2	NM	N/A	N/A	-61.3	57.9
Dropbox, Inc.	DBX	NasdaqGS	USD	27.32	9,502.0	3.1	30.7	16.0	24.38	N/A	-109.7	74.1
Dynatrace, Inc.	DT	NYSE	USD	47.60	13,961.0	0.4	41.5	45.0	29.91	N/A	9.4	N/A
NICE Ltd.	NICE	NasdaqGS	USD	158.24	10,074.0	-11.0	-16.5	858.0	N/A	N/A	10.1	11.5
Procore Technologies, Inc.	PCOR	NYSE	USD	64.55	9,191.0	2.2	24.4	NM	N/A	N/A	-22.9	N/A
Samsara Inc.	IOT	NYSE	USD	23.23	12,427.0	-14.4	102.9	NM	N/A	N/A	-25.4	N/A
Unity Software Inc.	U	NYSE	USD	27.49	10,541.0	-19.0	-6.8	NM	N/A	N/A	-33.1	41.4

*For Peer Groups with more than 10 companies or stocks, selection of issues is based on market capitalization.
NA-Not Available; NM-Not Meaningful.
Note: Peers are selected based on Global Industry Classification Standards and market capitalization. The peer group list includes companies with similar characteristics, but may not include all the companies within the same industry and/or that engage in the same line of business.

Manhattan Associates, Inc.

Analyst Research Notes and other Company News

July 26, 2023

02:01 PM ET... CFRA Maintains Buy Opinion on Shares of Manhattan Associates, Inc. [MANH 187.12****]:

We raise our target price to \$204 from \$186 on a P/E of 67x our next-12-month EPS estimate of \$3.05, in line with its three-year average on expanding margins. We increase our 2023 EPS projection to \$3.11 from \$2.95 and our 2024 EPS view to \$3.52 from \$3.29. MANH reported Q2 revenue of \$231M, above consensus by \$14.4M, and non-GAAP EPS of \$0.88 beat by \$0.16. Top-line results were led by Cloud Subscriptions growth [+44.3% Y/Y], driven by a combination of conversions, upsells, and cross-sells. Net new logos adds [about 25% of new cloud bookings], reflect solid demand for its cloud-native solutions and competitive win rates. The shift to cloud, however, compressed overall sales performance by approximately 700 bps, as expected, as Software License declined 28% Y/Y. We maintain a positive view on MANH as strong cloud uptake and operating leverage gains raise our EPS outlook for 2023 and 2024. We note that operating margins [non-GAAP] expanded 210 bps to 29.6% this quarter from Q2 of the prior year. / Janice Quek

April 26, 2023

12:34 PM ET... CFRA Downgrades Opinion on Manhattan Associates Inc. to a Buy from Strong Buy [MANH 169.58****]:

We cut our target price to \$186 from \$194, on a P/E of 66x our next-12-month EPS estimate of \$2.82, in line with its three-year average. We trim our 2023 EPS projection to \$2.95 from \$3.06 and our 2024 EPS view to \$3.29 from \$3.66. MANH reported Q1 revenue of \$221M, above consensus by \$20.5M, and non-GAAP EPS of \$0.80 beats by \$0.15. Top-line growth of 23.5% Y/Y was led by cloud subscriptions sales [an increase of 53.4% Y/Y] and professional services [+29% Y/Y]. Demand for MANH's products and win rates were robust, as its platform of unified mission-critical and supply chain solutions drove strong cross-selling momentum with its enterprise customers during the quarter. We note the relative resilience of the business even in a cautious spending environment. Raised guidance [+500 bps of top-line growth from February] reflect impressive execution and a solid outlook, but we see modest upside potential at current valuation multiples. / Janice Quek

February 06, 2023

12:21 AM ET... CFRA Reiterates Strong Buy Rating on Manhattan Associates, Inc. [MANH 149.81****]:

We raise our target by \$21 to \$194 based on: 1) strong Q4 '22 results, again, with 16% Y/Y revenue growth, driven mainly by cloud, which grew 49% Y/Y, a Q/Q acceleration from 41% growth in Q4, a very positive indicator of rapid customer adoption of MANH's new cloud offerings; 2) very impressive 22% Y/Y growth in professional services, up from 17% growth in Q3 and now 50% total, with 5% decline in legacy maintenance revenue, ~18% of total, so far pro services appear to have benefitted rather than been cannibalized from MANH's migration to the cloud, although there is a risk that the present surge in pro services falls off more rapidly as customers complete their cloud migration for Supply Chain Management [SCM] apps; 3) license revenue fell 58%, now less than 3% of total, no longer a drag on growth. Our target is the product of our \$3.66 EPS forecast for '24, now \$0.12 lower and pulled forward from '23, and a 53.0x P/E [6-month mean]. Our EPS forecast for '23 remains at \$3.06, and we initiate '25 at 4.40. / John Freeman

October 27, 2022

06:01 PM ET... CFRA Maintains Strong Buy Rating on Shares of Manhattan Associates, Inc. [MANH 118.11****]:

We cut our target by \$23 to \$173 based on: 1) strong 3Q22 results, in our view, with 17% Y/Y revenue growth, another Q/Q acceleration from 16% in 2Q and 14% in 1Q driven mainly by cloud [+41% Y/Y], and a deceleration from 48% in 2Q, but still faster than 1Q at 36% to 23% of total revenue; 2) healthy professional services growth of 17% Y/Y to 52% of total, with 4% legacy maintenance growth, an upside surprise for us, now 18% of total; 3) higher '22 midpoint guidance for revenue, now \$14m higher to \$752m, implying 13% Y/Y growth vs. 11% previously, and for EPS, now \$0.07 higher to \$2.44; and 4) more attractive valuation with little change in our fundamental view. Our target is the product of our \$3.06 EPS forecast for '23 [+ \$0.04] and a 56.6x P/E [6-month mean to better reflect present market conditions, down from 65.1x]. 3Q revenue of \$198m [+17% Y/Y] beat consensus by \$13m; EPS of \$0.66 declined \$0.05 Y/Y, beating by \$0.09. Our EPS for '22 remains \$2.56, but we lower our forecast for '24 by \$0.17 to \$3.78. / John Freeman

July 28, 2022

02:44 AM ET... CFRA Maintains Strong Buy Rating on Shares of Manhattan Associates, Inc. [MANH 137.44****]:

We trim our target by \$9 to \$196 based on: 1) strong 2Q22 results with 16% Y/Y revenue growth, a material acceleration from 14% growth in 1Q, driven mainly by cloud, up 48% Y/Y, a significant acceleration from 1Q growth of 36%, to 22% of total revenue; 2) professional services also accelerated, +19% Y/Y to 53% of total, partially offset by a 42% drop in legacy license revenue, now only 3% of total; 3) midpoint guidance for '22 revenue now \$14m higher to \$737m, now implying 11% Y/Y growth, and for EPS by \$0.19 to \$2.37. Our target is the product of our \$3.02 EPS forecast for '23, +\$0.06, and a 65.1x P/E [1-year mean, down from 69.1x]. 2Q revenue of \$192m, +16% Y/Y, beat consensus by \$12m; EPS of \$0.69, +\$0.08 Y/Y, beat by \$0.15. Our EPS forecast for '22 remains \$2.56, but we increase our forecast for '24 by \$0.07 to \$3.95. Our assumption of 3-year 30%+ cloud growth was bolstered again by strong 84% Y/Y higher Remaining Performance Obligation [RPO], a leading indicator for subscription revenue. / John Freeman

April 28, 2022

01:00 PM ET... CFRA Maintains Strong Buy Rating on Shares of Manhattan Associates, Inc. [MANH 128.81****]:

We trim our target by \$5 to \$205 based on: 1) solid 1Q21 results with 14% Y/Y revenue growth driven mainly by cloud subscriptions [+36% Y/Y to 21% of total] with professional services still healthy, up 12% Y/Y to 50% of total, partially offset by a 1% decline in legacy software license and maintenance revenue, 24% of total; and 2) impressive 420 bps Y/Y operating margin expansion to 27%. Our \$205 target is the product of our \$2.96 EPS forecast for '23, now \$0.16 lower, and a 69.1x P/E [2-year mean]. 4Q revenue of \$168m, +7% Y/Y, beat consensus by \$10m; EPS of \$0.48, +40% Y/Y, beat by \$0.15. We cut our EPS forecast for '22 by \$0.05 to \$2.56 but increase our forecast for '24 by \$0.16 to \$3.88. Management upped full-year midpoint guidance for revenue by \$16m [+8%] to \$223m and for EPS by \$0.14 [7%] to \$2.18. Our assumption of 30%+ cloud growth through 2023 is bolstered by another quarter of strong 92% Y/Y growth in Remaining Performance Obligation [RPO], a leading indicator for subscription revenue. / John Freeman

February 02, 2022

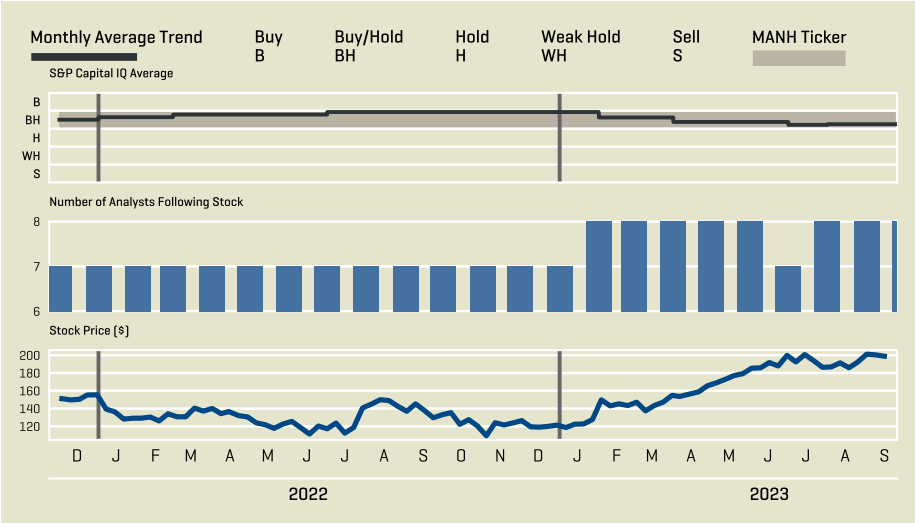
06:51 AM ET... CFRA Maintains Strong Buy Rating on Shares of Manhattan Associates, Inc. [MANH 135.22****]:

We cut our target by \$38 to \$210 given: 1) solid 4Q21 results with 16% Y/Y revenue growth driven mainly by cloud subscriptions [+48% Y/Y to 20% of total] and solid contributions from professional services [+15% Y/Y, 51% of total] and, surprisingly, legacy license sales [+24%, 7% of total], partially offset by a 4% Y/Y decline in legacy maintenance revenue, 22% of total; 2) strong 126% Y/Y growth in Remaining Performance Obligation [RPO], a leading indicator for recurring revenue, bolstering our conviction in the traction for the Manhattan Active cloud platform and cloud versions of its inventory, warehouse, and logistics apps; 3) somewhat tepid increases in midpoint guidance for '22 revenue [+ \$2m] and EPS [+ \$0.4]. Our \$210 target is the product of our \$3.12 EPS forecast for '23, now \$0.12 lower, and a 67.2x P/E [2-year mean]. 4Q revenue of \$171m, +16% Y/Y, beat consensus by \$9m; EPS of \$0.48, up \$0.03 Y/Y, beat by \$0.10. We also trim our EPS forecast for '22 by \$0.07 to \$2.61 and initiate '24 at \$3.72. / John Freeman

Note: Research notes reflect CFRA's published opinions and analysis on the stock at the time the note was published. The note reflects the views of the equity analyst as of the date and time indicated in the note, and may not reflect CFRA's current view on the company.

Manhattan Associates, Inc.

Analysts Recommendations



	No. of Recommendations	% of Total	1 Mo.Prior	3 Mos.Prior
Buy	2	25	2	2
Buy/Hold	2	25	2	1
Hold	4	50	4	4
Weak hold	0	0	0	0
Sell	0	0	0	0
No Opinion	0	0	0	0
Total	8	100	8	7

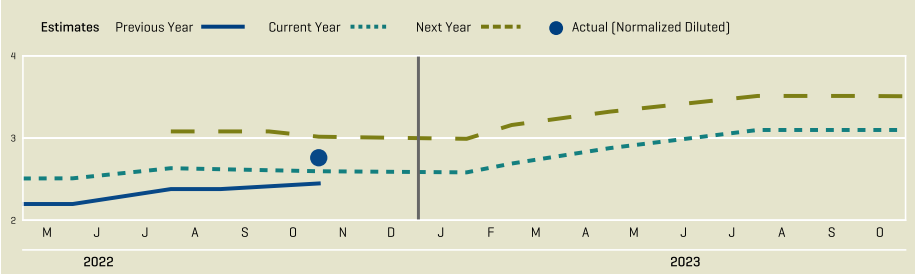
Wall Street Consensus Opinion

Buy/Hold

Wall Street Consensus vs. Performance

For fiscal year 2023, analysts estimate that MANH will earn USD 3.10. For fiscal year 2024, analysts estimate that MANH's earnings per share will grow by 13.24% to USD 3.51.

Wall Street Consensus Estimates



Fiscal Year	Avg Est.	High Est.	Low Est.	# of Est.	Est. P/E
2024	3.51	3.69	3.32	7	55.96
2023	3.10	3.12	3.06	7	63.37
2024 vs. 2023	▲ 13%	▲ 18%	▲ 9%	N/A%	▼ -12%
Q3'24	0.92	0.97	0.86	7	213.33
Q3'23	0.76	0.78	0.74	7	256.79
Q3'24 vs. Q3'23	▲ 20%	▲ 24%	▲ 16%	N/A%	▼ -17%

Forecasts are not reliable indicator of future performance.
Note: A company's earnings outlook plays a major part in any investment decision. S&P Global Market Intelligence organizes the earnings estimates of over 2,300 Wall Street analysts, and provides their consensus of earnings over the next two years, as well as how those earnings estimates have changed over time. Note that the information provided in relation to consensus estimates is not intended to predict actual results and should not be taken as a reliable indicator of future performance.
Note: For all tables, graphs and charts in this report that do not cite any reference or source, the source is S&P Global Market Intelligence.

Manhattan Associates, Inc.

Glossary

STARS

Since January 1, 1987, CFRA Equity and Fund Research Services, and its predecessor S&P Capital IQ Equity Research has ranked a universe of U.S. common stocks, ADRs [American Depositary Receipts], and ADSs [American Depositary Shares] based on a given equity's potential for future performance. Similarly, we have ranked Asian and European equities since June 30, 2002. Under proprietary STARS [Stock Appreciation Ranking System], equity analysts rank equities according to their individual forecast of an equity's future total return potential versus the expected total return of a relevant benchmark [e.g., a regional index (MSCI AC Asia Pacific Index, MSCI AC Europe Index or S&P 500® Index)], based on a 12-month time horizon. STARS was designed to help investors looking to put their investment decisions in perspective. Data used to assist in determining the STARS ranking may be the result of the analyst's own models as well as internal proprietary models resulting from dynamic data inputs.

S&P Global Market Intelligence's Quality Ranking

[also known as **S&P Capital IQ Earnings & Dividend Rankings**] - Growth and S&P Capital IQ Earnings & Dividend Rankings stability of earnings and dividends are deemed key elements in establishing S&P Global Market Intelligence's earnings and dividend rankings for common stocks, which are designed to capsule the nature of this record in a single symbol. It should be noted, however, that the process also takes into consideration certain adjustments and modifications deemed desirable in establishing such rankings. The final score for each stock is measured against a scoring matrix determined by analysis of the scores of a large and representative sample of stocks. The range of scores in the array of this sample has been aligned with the following ladder of rankings:

A+ Highest	B Below Average
A High	B- Lower
A Above	C Lowest
B+ Average	D In Reorganization
NC Not Ranked	

EPS Estimates

CFRA's earnings per share (EPS) estimates reflect analyst projections of future EPS from continuing operations, and generally exclude various items that are viewed as special, non-recurring, or extraordinary. Also, EPS estimates reflect either forecasts of equity analysts; or, the consensus [average] EPS estimate, which are independently compiled by S&P Global Market Intelligence, a data provider to CFRA. Among the items typically excluded from EPS estimates are asset sale gains; impairment, restructuring or merger-related charges; legal and insurance settlements; in process research and development expenses; gains or losses on the extinguishment of debt; the cumulative effect of accounting changes; and earnings related to operations that have been classified by the company as discontinued. The inclusion of some items, such as stock option expense and recurring types of other charges, may vary, and depend on such factors as industry practice, analyst judgment, and the extent to which some types of data is disclosed by companies.

12-Month Target Price

The equity analyst's projection of the market price a given security will command 12 months hence, based on a combination of intrinsic, relative, and private market valuation metrics, including Fair Value.

Abbreviations Used in Equity Research Reports

- CAGR - Compound Annual Growth Rate
- CAPEX - Capital Expenditures
- CY - Calendar Year
- DCF - Discounted Cash Flow
- DDM - Dividend Discount Model
- EBIT - Earnings Before Interest and Taxes
- EBITDA - Earnings Before Interest, Taxes, Depreciation & Amortization
- EPS - Earnings Per Share
- EV - Enterprise Value
- FCF - Free Cash Flow
- FFO - Funds From Operations
- FY - Fiscal Year
- P/E - Price/Earnings
- P/NAV - Price to Net Asset Value
- PEG Ratio - P/E-to-Growth Ratio
- PV - Present Value
- R&D - Research & Development
- ROCE - Return on Capital Employed
- ROE Return on Equity
- ROI - Return on Investment
- ROIC - Return on Invested Capital
- ROA - Return on Assets
- SG&A - Selling, General & Administrative Expenses
- SOTP - Sum-of-The-Parts
- WACC - Weighted Average Cost of Capital

Dividends on American Depositary Receipts (ADRs) and American Depositary Shares (ADSs) are net of taxes (paid in the country of origin).

Qualitative Risk Assessment

Reflects an equity analyst's view of a given company's operational risk, or the risk of a firm's ability to continue as an ongoing concern. The Qualitative Risk Assessment is a relative ranking to the U.S. STARS universe, and should be reflective of risk factors related to a company's operations, as opposed to risk and volatility measures associated with share prices. For an ETF this reflects on a capitalization-weighted basis, the average qualitative risk assessment assigned to holdings of the fund.

STARS Ranking system and definition:

★★★★★ 5-STARS (Strong Buy):

Total return is expected to outperform the total return of a relevant benchmark, by a notable margin over the coming 12 months, with shares rising in price on an absolute basis.

★★★★★ 4-STARS (Buy):

Total return is expected to outperform the total return of a relevant benchmark over the coming 12 months.

★★★★★ 3-STARS (Hold):

Total return is expected to closely approximate the total return of a relevant benchmark over the coming 12 months.

★★★★★ 2-STARS (Sell):

Total return is expected to underperform the total return of a relevant benchmark over the coming 12 months.

★★★★★ 1-STAR (Strong Sell):

Total return is expected to underperform the total return of a relevant benchmark by a notable margin over the coming 12 months, with shares falling in price on an absolute basis.

Relevant benchmarks:

In North America, the relevant benchmark is the S&P 500 Index, in Europe and in Asia, the relevant benchmarks are the MSCI AC Europe Index and the MSCI AC Asia Pacific Index, respectively.



Manhattan Associates, Inc.

Disclosures

Stocks are ranked in accordance with the following ranking methodologies:

STARS Stock Reports:

Qualitative STARS rankings are determined and assigned by equity analysts. For reports containing STARS rankings refer to the Glossary section of the report for detailed methodology and the definition of STARS rankings.

Quantitative Stock Reports:

Quantitative rankings are determined by ranking a universe of common stocks based on 5 measures or model categories: Valuation, Quality, Growth, Street Sentiment, and Price Momentum. In the U.S., a sixth sub-category for Financial Health will also be displayed. Percentile scores are used to compare each company to all other companies in the same universe for each model category. The five [six] model category scores are then weighted and rolled up into a single percentile ranking for that company. For reports containing quantitative rankings refer to the Glossary section seof the report for detailed methodology and the definition of Quantitative rankings.

STARS Stock Reports and Quantitative Stock Reports:

The methodologies used in STARS Stock Reports and Quantitative Stock Reports [collectively, the "Research Reports"] reflect different criteria, assumptions and analytical methods and may have differing rankings. The methodologies and data used to generate the different types of Research Reports are believed by the author and distributor reasonable and appropriate. Generally, CFRA does not generate reports with different ranking methodologies for the same issuer. However, in the event that different methodologies or data are used on the analysis of an issuer, the methodologies may lead to different views on the issuer, which may at times result in contradicting assessments of an issuer. CFRA reserves the right to alter, replace or vary models, methodologies or assumptions from time to time and without notice to clients.

STARS Stock Reports:

Global STARS Distribution as of September 30, 2023

Ranking	North America	Europe	Asia	Global
Buy	36.8%	34.5%	41.6%	37.4%
Hold	53.8%	51.2%	50.3%	52.6%
Sell	9.4%	14.3%	8.1%	10.1%
Total	100.0%	100.0%	100.0%	100.0%

Analyst Certification:

STARS Stock Reports are prepared by the equity research analysts of CFRA and its affiliates and subsidiaries. Quantitative Stock Reports are prepared by CFRA. All of the views expressed in STARS Stock Reports accurately reflect the research analyst's personal views regarding any and all of the subject securities or issuers; all of the views expressed in the Quantitative Stock Reports accurately reflect the output of CFRA's algorithms and programs. Analysts generally update STARS Stock Reports at least four times each year. Quantitative Stock Reports are generally updated weekly. No part of analysts' or CFRA's compensation was, is, or will be directly or indirectly related to the specific rankings or views expressed in any Stock Report.

About CFRA Equity Research:

This Research Report is published and originally distributed by Accounting Research & Analytics, LLC d/b/a CFRA ["CFRA US"], with the following exceptions: In the UK/EU/EEA, it is published and originally distributed by CFRA UK Limited ["CFRA UK"], which is regulated by the Financial Conduct Authority [No. 775151], and in Malaysia by CFRA MY Sdn Bhd [Company No. 683377-A] ["CFRA Malaysia"], which is regulated by Securities Commission Malaysia, [No. CMSL/A0181/2007] under license from CFRA US. These parties and their subsidiaries maintain no responsibility for reports redistributed by third parties such as brokers or financial advisors.

General Disclosure

Notice to all jurisdictions:

Where Research Reports are made available in a language other than English and in the case of inconsistencies between the English and translated versions of a Research Report, the English version will control and supersede any ambiguities between such versions. Neither CFRA nor its affiliates guarantee the accuracy of any translation.

The content of this report and the opinions expressed herein are those of CFRA based upon publicly-available information that CFRA believes to be reliable and the opinions are subject to change without notice. This analysis has not been submitted to, nor received approval from, the United States Securities and Exchange Commission or any other regulatory body. CFRA AND ALL RELATED ENTITIES SPECIFICALLY DISCLAIM ALL WARRANTIES, EXPRESS OR IMPLIED, to the full extent permitted by law, regarding the accuracy, completeness, or usefulness of this information and assumes no liability with respect to the consequences of relying on this information for investment or other purposes.

No content in this Research Report may be modified, reverse engineered, reproduced or distributed in any form by any means, or stored in a database or retrieval system, without the prior written permission of CFRA, or used for any unlawful or unauthorized purposes. Neither CFRA nor its third-party providers, as well as its/their directors, officers, shareholders, employees or agents, guarantee the accuracy, completeness, timeliness or availability of the content herein

Past performance is not necessarily indicative of future results.

This document may contain forward-looking statements or forecasts; such forecasts are

not a reliable indicator of future performance.

This report is not intended to, and does not, constitute an offer or solicitation to buy and sell securities or engage in any investment activity. This report is for informational purposes only. Statements in this report are not made with respect to any particular investor or type of investor. Securities, financial instruments or strategies mentioned herein may not be suitable for all investors and this material is not intended for any specific investor and does not take into account an investor's particular investment objectives, financial situations or needs. Before acting on anything in this report, you should consider whether it is suitable for your particular circumstances and, if necessary, seek professional advice. CFRA may license certain intellectual property or provide services to, or otherwise have a business relationship with, certain issuers of securities that are the subject of CFRA research reports, including exchange-traded investments whose investment objective is to substantially replicate the returns of a proprietary index of CFRA. In cases where CFRA is paid fees that are tied to the amount of assets invested in a fund or the volume of trading activity in a fund, investment in the fund may result in CFRA receiving compensation in addition to the subscription fees or other compensation for services rendered by CFRA, however, no part of CFRA's compensation for services is tied to any particular viewpoint or rating. Additional information on a subject company may be available upon request.

CFRA's financial data provider is S&P Global Market Intelligence. THIS DOCUMENT CONTAINS COPYRIGHTED AND TRADE SECRET MATERIAL DISTRIBUTED UNDER LICENSE FROM S&P GLOBAL MARKET INTELLIGENCE. FOR RECIPIENT'S INTERNAL USE ONLY.

The Global Industry Classification Standard [GICS®] was developed by and/or is the exclusive property of MSCI, Inc. and S&P Global Market Intelligence. GICS is a service mark of MSCI and S&P Global Market Intelligence and has been licensed for use by CFRA.

Other Disclaimers and Notices

Certain information in this report is provided by S&P Global, Inc. and/or its affiliates and subsidiaries [collectively "S&P Global"]. Such information is subject to the following disclaimers and notices: "Copyright © 2018, S&P Global Market Intelligence [and its affiliates as applicable]. All rights reserved. Nothing contained herein is investment advice and a reference to a particular investment or security, a credit rating or any observation concerning a security or investment provided by S&P Global is not a recommendation to buy, sell or hold such investment or security or make any other investment decisions. This may contain information obtained from third parties, including ratings from credit ratings agencies. Reproduction and distribution of S&P Global's information and third party content in any form is prohibited except with the prior written permission of S&P Global or the related third party, as applicable. Neither S&P Global nor its third party providers guarantee the accuracy, completeness, timeliness or availability of any information, including ratings, and are not responsible for any errors or omissions [negligent or otherwise], regardless of the cause, or for the results obtained from the use of such information or content. S&P GLOBAL AND ITS THIRD PARTY CONTENT PROVIDERS GIVE NO EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, ANY WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE OR USE AND ALL S&P INFORMATION IS PROVIDED ON AN AS-IS BASIS. S&P GLOBAL AND ITS THIRD PARTY CONTENT PROVIDERS SHALL NOT BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, EXEMPLARY, COMPENSATORY, PUNITIVE, SPECIAL OR CONSEQUENTIAL DAMAGES, COSTS, EXPENSES, LEGAL FEES, OR LOSSES [INCLUDING LOST INCOME OR PROFITS AND OPPORTUNITY COSTS OR LOSSES CAUSED BY NEGLIGENCE] IN CONNECTION WITH ANY USE OF THEIR INFORMATION OR CONTENT, INCLUDING RATINGS. Credit ratings are statements of opinions and are not statements of fact or recommendations to purchase, hold or sell securities. They do not address the suitability of securities or the suitability of securities for investment purposes, and should not be relied on as investment advice."

Manhattan Associates, Inc.

CFRA's Research Reports may be distributed in certain localities, countries and/or jurisdictions by independent third parties or independent intermediaries and/or distributors ("Intermediaries"). Intermediaries are not acting as agents or representatives of CFRA. In territories where an Intermediary distributes CFRA's Research Reports, the Intermediary, and not CFRA, is solely responsible for complying with all applicable regulations, laws, rules, circulars, codes and guidelines established by local and/or regional regulatory authorities, including laws in connection with the distribution of third party research reports, licensing requirements, supervisory and record keeping obligations that the Intermediary may have under the applicable laws and regulations of the territories where it distributes the Research Reports.

For residents of the European Union/European Economic Area:

Research reports are originally distributed by CFRA UK Limited (company number 08456139 registered in England & Wales with its registered office address at New Derwent House, 69-73 Theobalds Road, London, WC1X 8TA, United Kingdom). CFRA UK Limited is regulated by the UK Financial Conduct Authority (No. 775151).

For residents of Malaysia:

Research reports are originally produced and distributed by CFRA MY Sdn Bhd (Company No. 683377-A) ("CFRA Malaysia"), a wholly-owned subsidiary of CFRA US. CFRA Malaysia is regulated by Securities Commission Malaysia (License No. CMSL/A0181/2007).

For Recipients in Canada:

This report is not prepared subject to Canadian disclosure requirements and may not be suitable for Canadian investors.

For residents of Singapore:

This Research Report is distributed by CFRA UK Limited to its clients in Singapore who hold a financial advisers licence or is a person exempt from holding such licence ("SG Intermediary"). Recipients of this Research Report in Singapore should contact the SG Intermediary in respect to any matters arising from, or in connection with, the analysis in this report. Where the recipient is not an accredited, expert or institutional investor as defined by the Securities and Futures Act, the SG Intermediary accepts legal responsibility for the contents of this Research Report in accordance with applicable law. When reports are distributed by SG Intermediaries in Singapore, the SG Intermediary, and not CFRA, is solely responsible for ensuring that the recipients of the Research Reports understand the information contained in the Research Reports and that such information is suitable based on the customer's profile and investment objectives. This Research Report is intended for general circulation and no advice or recommendation is made herein or by CFRA to any particular person. CFRA does not assume any responsibility to advise on whether any particular product is suitable for any person, and the analysis herein does not take into account the specific investment objectives, financial situation or particular needs of any particular person, and should not be relied upon for any investment decision.

For residents of all other countries:

Research reports are originally distributed Accounting Research & Analytics, LLC d/b/a CFRA.

Copyright © 2023 CFRA. All rights reserved. CFRA and STARS are registered trademarks of CFRA.

EFG Disclaimer

The attached document is produced for it to be used by entities, including affiliates and branches, of EFG International AG. The information herein is provided on the basis that it is a suitable investment for the recipient taking into account the recipient's personal circumstances and should therefore be considered as a personal recommendation. Tax implications should be nevertheless independently assessed by the recipient with a tax advisor before entering into any transaction.

The information provided herein is for the exclusive use of the recipient and may not be reproduced, disclosed or distributed, neither in part nor in full.

Investment products may be subject to investment risks involving, but not limited to, possible loss of all or part of the principal invested.

The information and views expressed herein at the time of writing are subject to change at any time without notice and there is no obligation to update or remove outdated information. Historical data on the performance of the securities and financial instruments or the underlying assets in this document is no indication for future performance and the value of investments may fall as well as rise. This document is not directed at, or intended for distribution to or use by, any person or entity domiciled or resident in any jurisdiction where such distribution, publication, availability or use would be contrary to applicable laws or regulations of such jurisdictions.

EFG and its employees may engage in securities transactions, on a proprietary basis or otherwise and hold long or short positions with regard to the instruments identified herein; such transactions or positions may be inconsistent with the views expressed in the document.

If you have received this document from any affiliate or branch referred to below, please note the following:

Switzerland: EFG Bank AG, Zurich, including its Geneva and Lugano branches, is authorised and regulated by the FINMA. Registered Office: EFG Bank AG, Bleicherweg 8, 8001 Zurich, Switzerland. Registered Swiss Branches: EFG Bank SA, 24 quai du Seujet, 1211 Geneva 2, and EFG Bank SA, Via Magatti 2, 6900 Lugano.

The information provided in this document is not the result of a financial research conducted by the Bank's research department. Therefore, the "Directive on the Independence of Financial Research" issued by the Swiss Banking Association does not apply to it. As the document is not intended to be exhaustive on all risks related to financial instruments, the recipient should consult the specific product documentation and the June 2023 "Risks Involved in Trading Financial Instruments" brochure released by the Swiss Bankers Association and available on the Swiss Bankers Association website:

<https://www.swissbanking.org/en/services/library/guidelines> or free of charge from EFG Bank AG.

Luxembourg: EFG Bank (Luxembourg) S.A. is authorised by the Ministry of Finance Luxembourg and supervised by the Commission de Surveillance du Secteur Financier (CSSF). EFG Bank (Luxembourg) S.A. is Member of the Deposit Guarantee Fund Luxembourg (F.G.D.L. - Fonds de Garantie des Dépôts Luxembourg) and Member of the Luxembourg Investor Compensation Scheme (S.I.I.L. - Système d'Indemnisation des Investisseurs Luxembourg). R.C.S. Luxembourg no. B113375. Registered address: EFG Bank (Luxembourg) S.A. - 56, Grand-Rue, L-1660 Luxembourg.

Portugal: EFG Bank (Luxembourg) S.A. - Sucursal em Portugal is authorised and supervised by Banco de Portugal (register 280) and the CMVM, the Portuguese securities market commission, (register 393) for the provision of financial advisory and reception and transmission of orders. EFG Bank (Luxembourg) S.A. - Sucursal em Portugal is a non-booking branch of EFG Bank (Luxembourg) S.A., a public limited liability company incorporated under the laws of the Grand Duchy of Luxembourg, authorised and supervised by the CSSF (Commission de Surveillance du Secteur Financier). Lisbon Head Office: Avenida da Liberdade n.º 131 - 6º Dto., 1250 - 140 Lisboa. Porto agency: Avenida da Boavista, n.º 1837 - Escritório 6.2, 4100-133 Porto. Companies Registry Number: 980649439.

Greece: EFG Bank (Luxembourg) S.A., Athens Branch is an non-booking establishment of EFG Bank (Luxembourg) S.A. which is authorised to promote EFG Bank (Luxembourg) S.A.'s Greece: EFG Bank (Luxembourg) S.A., Athens Branch is an non-booking establishment of EFG Bank (Luxembourg) S.A. which is authorised to promote EFG Bank (Luxembourg) S.A.'s products and services based on the EU freedom of establishment pursuant to a license granted by the Luxembourg financial supervisory authority "CSSF". Registered address: 342 Kifisias Ave. & Ethnikis Antistaseos Str. - 154 51 N. Psychiko, General Commercial Registry no. 143057760001.

Liechtenstein: EFG Bank von Ernst AG is regulated by the Financial Market Authority Liechtenstein. Registered address: EFG Bank von Ernst AG Egertastrasse 10 - 9490 Vaduz, Liechtenstein.

Monaco: EFG Bank (Monaco) SAM is a Monegasque Public Limited Company with a company registration no. 90 S 02647 (Registre du Commerce et de l'Industrie de la Principauté de Monaco). EFG Bank (Monaco) SAM is a bank with financial activities authorised and regulated by the French Prudential Supervision and Resolution Authority and by the Monegasque Commission for the Control of Financial Activities. Registered address: EFG Bank (Monaco) SAM, Villa les Aigles, 15, avenue d'Ostende - BP 37 - 98001 Monaco (Principauté de Monaco), telephone: +377 93 15 11 11. The recipient of this document is perfectly fluent in English and waives the possibility to obtain a French version of this publication.

United Kingdom: EFG Private Bank Limited is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority. EFG Private Bank Limited is a member of the London Stock Exchange. Registered company no. 2321802. Registered address: EFG Private Bank Limited, Park House, 116 Park Street, London W1K 6AP, United Kingdom, telephone +44 (0)20 7491 9111.

Dubai: EFG (Middle East) Limited is regulated by the DFSA. This material is intended "for professional clients only". Registered address: EFG (Middle East) Limited DIFC, Gate Precinct 5, 7th Floor PO Box 507245 - Dubai, UAE.